

Lecture 9 Team Projects

Interest-Rate Futures

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1 Project 1: Hedge a Bond Position with Treasury Futures

Scenario. Reduce a bond position's one-month rate exposure without selling the bonds.

Objectives. Estimate DV01; choose a Treasury futures contract and hedge ratio; explain conversion factors, cheapest-to-deliver uncertainty, basis risk, and slippage.

Sources. CME Treasury futures specifications and settlements; U.S. Treasury and FINRA data; instructor delivery-basket snapshot.

Deliverables. Position risk, contract choice, hedge ratio, two curve scenarios, basis-risk note, and a 10–12 minute presentation.

Rubric (100 points). Risk measurement 20; hedge ratio 30; delivery and basis analysis 25; scenarios 15; presentation and sources 10.

2 Project 2: Read the SOFR Futures Curve

Scenario. Brief a desk on the policy path implied by SOFR futures around a central-bank announcement.

Objectives. Convert prices to implied rates; compare strips on two dates; distinguish timing from magnitude; explain why implied rates are not pure forecasts.

Sources. CME SOFR settlements; Federal Reserve releases; FRED; instructor settlement snapshot.

Deliverables. Two strips, change table, event timeline, risk-premium caveats, desk conclusion, and a 10–12 minute presentation.

Rubric (100 points). Conversion 25; curve analysis 25; policy interpretation 20; caveats 20; presentation 10.

3 Project 3: Cheapest-to-Deliver Switch Risk

Scenario. Assess whether a futures hedge remains reliable when the likely cheapest-to-deliver Treasury changes.

Objectives. Compare delivery candidates, conversion-factor-adjusted prices, and implied repo rates; identify switch conditions; measure hedge-ratio sensitivity.

Sources. CME delivery rules and conversion factors; U.S. Treasury terms; instructor cash/futures quote snapshot.

Deliverables. Delivery-basket table, CTD calculation, two switch scenarios, revised hedge ratios, recommendation, and a 10–12 minute presentation.

Rubric (100 points). Contract mechanics 20; CTD analysis 30; switch scenarios 20; hedge implications 20; presentation and auditability 10.

4 Project 4: Duration Overlay for a Bond Fund

Scenario. Temporarily move a bond fund's duration to a tactical target while preserving its cash holdings.

Objectives. Translate target duration into DV01; select one or two futures contracts; respect cash, margin, and position constraints; define unwind triggers.

Sources. Public fund holdings and fact sheet; CME specifications; U.S. Treasury yields; instructor margin assumptions.

Deliverables. Starting exposure, overlay trades, after-trade duration, margin and stress table, governance plan, and a 10–12 minute presentation.

Rubric (100 points). Exposure diagnosis 20; overlay construction 30; constraints 20; governance and risk 20; presentation and sources 10.